A Wave in a Grid · AS · Practice

U09c · 24 written + 6 pen-first = 30 marks · Name: _____ Due: _____

Fresh authored questions. Show working. Unless stated otherwise, calculate packed, uncompressed PCM sample data only: no header, padding or compression. 1 byte = 8 bits; 1 kHz = 1000 Hz; 1 KiB = 1024 bytes; 1 MiB = 1024 KiB.

For decimal units: 1 MB = 1 000 000 bytes.

1. Ours: from voltage to codes

(a) Distinguish analogue data from digital data. **[2]**

(b) Explain what is meant by quantisation error. **[2]**

2. Ours: the unit chooses the divisor

(a) A mono recording uses 48 kHz, 24 bits per sample and 40 seconds. Calculate its sample-data size in MB. Show working. **[2]**

(b) A stereo recording has 2 channels, 32 000 Hz, 16 bits per sample and 16 seconds. Calculate its sample-data size in MiB. Show working. **[2]**

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3. Ours: explain the conversion

Describe how an analogue signal from a microphone is converted into stored digital sound data. **[3]**

4. Ours: different dials, different errors

(a) For a new recording, sample resolution increases from 8 to 16 bits, with rate, channels and duration unchanged. State the size factor and explain one potential accuracy benefit. **[3]**

(b) Instead, sample rate rises while resolution stays fixed. Explain why this reduces the time gaps between samples but does not change the amplitude grid's step size. **[2]**

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5. Ours: count what is stored

(a) A mono recording stores 20 000 samples per second, using **2 bytes per sample**, for 30 seconds. Calculate its sample-data size in MB. Show working. **[2]**

(b) Its file contains a 256-byte header followed by exactly those raw sample bytes, with no padding or other data. Calculate the complete file size in bytes. State one item the header may store to interpret the samples. **[2]**

6. Ours: new information or more stored data?

(a) A recording lasts twice as long and has two channels instead of one. Rate and resolution are unchanged. State the size multiplier and explain whether these changes increase amplitude precision. **[2]**

(b) Explain why converting an existing low-resolution recording to more bits per sample cannot by itself identify amplitude differences already discarded. **[2]**

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7. Ours: build a tiny sound encoder · pen first [6]

💻 **Design, then trace. No code is required.** An encoder measures 4 samples per second. Its 3-bit codes represent amplitude levels 0 to 7. Round each input to its nearest level, then store that level as a three-bit unsigned binary code. The four inputs, in time order, are **0.2, 2.8, 5.1, 6.9**. Human marked.

(a) Write numbered steps for turning the supplied amplitude measurements into stored codes. Include the rounding rule, code width and order. **[2]**

(b) Trace your steps. Write the four rounded levels and the corresponding three-bit codes, in time order. **[2]**

(c) Using the same rate and resolution, a mono recording lasts 2 seconds. Calculate the number of samples and the packed sample-data size in bytes. **[2]**

Above and beyond · outside the 30 marks. A steady amplitude is 3.2. Two experimental encoders output four integer levels: A gives **3, 3, 3, 3**; B gives **3, 3, 3, 4**. Calculate each sequence's mean. Which mean is closer to 3.2? Does that mean every individual sample in B is more accurate? Explain.

Additional space - label the question number.